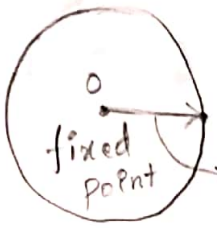
Circular Motion

- Horizontal
- Vertical

Horizontal Circular Motion



AOR = Axis of rotation
↳ [line passing through fixed point & ⊥ler to plane of motion]

*→ **Basic definitions:-**

① **Angular displacement (θ):**

* Angle sweeps out by radius vector.
* It is measured in "rad".
* For one complete rotation $\Rightarrow \theta = 2\pi \text{ rad} = 360^\circ$
* $1 \text{ rad} = \frac{180^\circ}{\pi}$, $1^\circ = \frac{\pi}{180} \text{ rad}$.

* **Points:**

* **Finite angular displacements** are not vectors but are "scalars" because they donot obey commutative law. i.e, $\bar{\theta}_1 + \bar{\theta}_2 \neq \bar{\theta}_2 + \bar{\theta}_1$.
* **Infinitesimal angular displacements** are not scalars but are "vectors".

$$d\bar{\theta}_1 + d\bar{\theta}_2 = d\bar{\theta}_2 + d\bar{\theta}_1$$

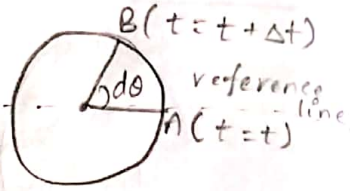
* Direction of angular displacement is always along the axis of rotation and is decided by right hand thumb rule.



1) Angular Velocity: (ω)

Rate of change of angular displacement.

① Avg angular velocity: ($\langle \omega \rangle$)



$$\text{Avg } \omega = \langle \omega \rangle = \frac{\Delta\theta}{\Delta t}$$

$$\boxed{\langle \omega \rangle = \frac{\theta_2 - \theta_1}{t_2 - t_1} \text{ rad/sec}}$$

② Instantaneous angular velocity:

$$\omega = \lim_{\Delta t \to 0} \left(\frac{\Delta\theta}{\Delta t} \right) = \frac{d\theta}{dt}$$

$$\boxed{\omega_{inst} = d\theta/dt}$$

*Points:

* ω is an axial vector. It's direction is along axis of rotation & is decided by right hand thumb ~~rule~~ rule.



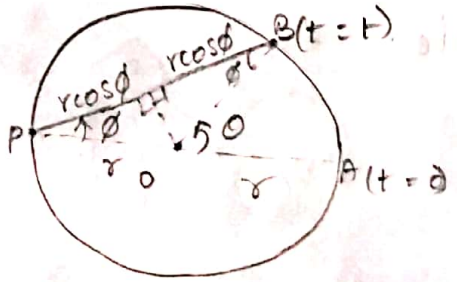
ω, θ → same direction
↳ outward



ω, θ → same direction
↳ inward

* It depends on axis of rotation i.e, if ACR is changed then 'ω' same particle will change.

* For ex;



$$\omega_0 = \frac{\theta}{t}$$

$$\omega_p = \frac{\phi}{t}$$

From figure,

$$\theta = 2\phi$$

$$\frac{\theta}{t} = 2\frac{\phi}{t}$$

$$\boxed{\omega_0 = 2\omega_p}$$ *

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** For uniform circular motion, $\boxed{\omega = \text{const}}$

i.e, $\boxed{\langle \omega \rangle = \omega_{inst}}$

* $\omega = \frac{\theta}{t}$ for; 1 complete rotation

$\theta \to 2\pi \Rightarrow t \to T \to$ time period of revolution

$$\boxed{\omega = \frac{2\pi}{T}}^{*}$$

we have $f = \frac{1}{T}$ → frequency of revolution (Hz / rps / s^{-1})

↳ no. of revolutions in 1 sec.

$$\boxed{\omega = 2\pi f}^{*}$$

↳ also called Angular frequency.

* 60 rpm = 1 rps

* 3600 rph = 1 rps

$\Rightarrow$ 1 rpm $= \frac{1}{60}$ rps

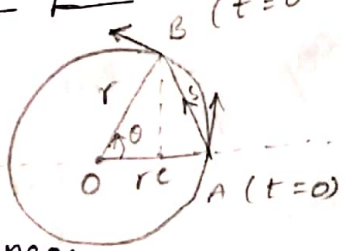
1 rph $= \frac{1}{3600 \times 12}$ rps

→ For x rpm $\Rightarrow$ $\omega = \frac{2\pi}{60} \times x$ rad/sec

For x rph $\Rightarrow$ $\omega = \frac{2\pi}{3600 \times 12} \times x$ rad/sec

Important points :-

*



$\bar{v} = v_x\hat{i} + v_y\hat{j}$

$|\bar{v}| = \sqrt{v_x^2 + v_y^2}$

distance:

dis = speed × time $= (\sqrt{v_x^2 + v_y^2}) \times t$

(or) distance = arc length (AB)

$$\boxed{AB = r\theta}^{*}$$

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Avg speed $= \frac{D}{t} = \frac{r\theta}{t}$

displacement (s):

From figure, $OC = r\cos\theta$, $BC = r\sin\theta$.

$AC = OA - OC$

$= r - r\cos\theta = r(1-\cos\theta)$

$S = (AC)^2 + (BC)^2$

$= r\sqrt{(1-\cos\theta)^2 + \sin^2\theta}$

$= r\sqrt{2 - 2\cos\theta}$

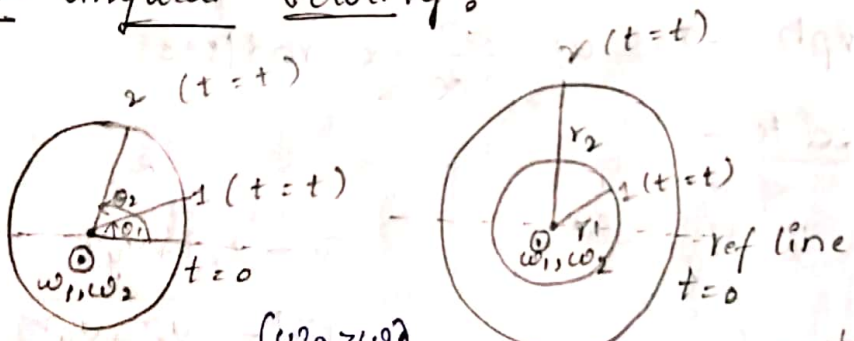
$= r\sqrt{4\sin^2\theta/2}$

$\boxed{S = 2r\sin\theta/2}$ ✓ here $\boxed{\theta = \omega t}$

Avg velocity $= \frac{S}{t}$

$= \frac{2r\sin\theta/2}{t}$

Relative angular velocity:



clearly, $\theta = \theta_2 - \theta_1$

$\Rightarrow \frac{\theta}{t} = \frac{\theta_2}{t} - \frac{\theta_1}{t}$

$\omega_{21} = \omega_2 - \omega_1$

$\frac{2\pi}{T} = \frac{2\pi}{T_2} - \frac{2\pi}{T_1} \Rightarrow \boxed{T = \frac{T_1 T_2}{T_1 - T_2}}$ *

↑ time taken by particles to complete one revolution w.r.t particle 1.

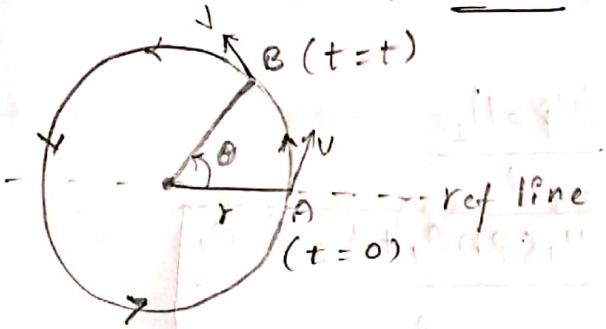
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* If both are moving in opposite directions.

$$\omega_{rel} = \omega_1 + \omega_2$$

$$\boxed{T = \frac{T_1 T_2}{T_1 + T_2}}$$ * → Time taken by one particle to complete one revolution with r.to other.

* Relation b/w v, r, ω :



Arc length

$AB = v(\Delta t)$

we have,

$$\text{Angle} = \frac{AB}{r}$$

$$\theta = \frac{v(\Delta t)}{r}$$

$$\frac{\theta}{\Delta t}(r) = v \Rightarrow \boxed{v = r\omega}$$ *

if ω — const

↳ Uniform circular motion.

* In vector.

$$\boxed{\bar{v} = \bar{\omega} \times \bar{r}}$$ *



clearly, $\boxed{\bar{v} \perp \bar{\omega} \perp \bar{r}}$

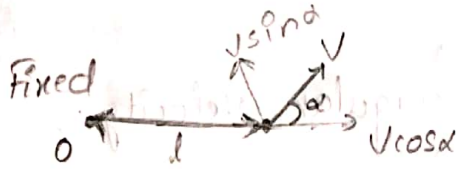
$\bar{v} \cdot \bar{\omega} = 0$

$\bar{v} \cdot \bar{r} = 0$

$\bar{\omega} \cdot \bar{r} = 0$

} dot product

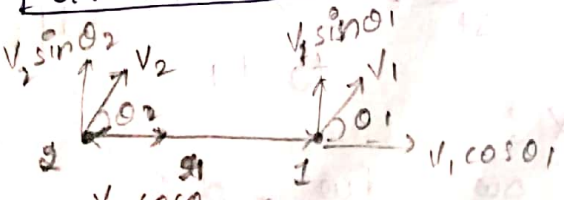
*



$$\boxed{\omega_o = \frac{v\sin\alpha}{l}}$$

& $\boxed{\frac{dl}{dt} = v\cos\alpha}$

*



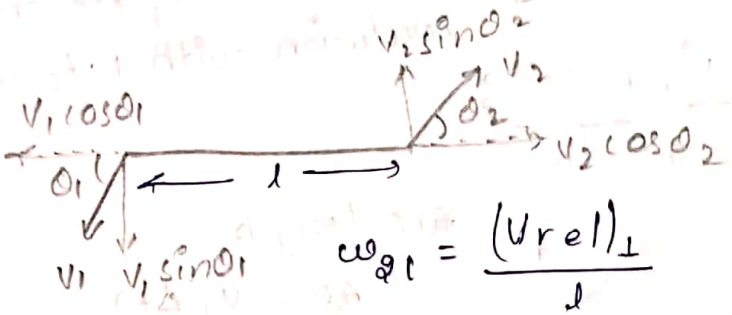
relative angular velocity, $\omega_{21} = \frac{(v_{rel})_1}{l}$

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$$\omega_{21} = \frac{V_2 \sin\theta_2 - V_1 \sin\theta_1}{r}$$

& $\frac{dl}{dt} = V_2 \cos\theta_2 - V_1 \cos\theta_1$

$\Rightarrow \frac{dl}{dt} = 0 \Rightarrow V_2 \cos\theta_2 = V_1 \cos\theta_1$ *

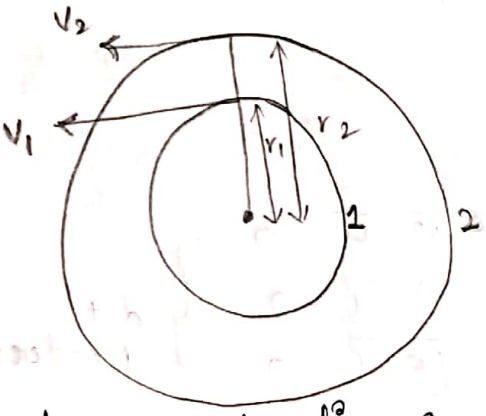


$$\omega_{21} = \frac{(V_{rel})_\perp}{l}$$

$$\omega_{21} = \frac{V_1 \sin\theta_1 + V_2 \sin\theta_2}{l}$$

& $\frac{dl}{dt} = V_1 \cos\theta_1 + V_2 \cos\theta_2$.

→



$$\omega_{21} = \frac{(V_{rel})_\perp}{r_{rel}} = \frac{V_2 - V_1}{r_2 - r_1}$$

③ Angular acceleration (α):

Rate of change of angular velocity.

① Avg α :

$$\langle \bar{\alpha} \rangle = \frac{\Delta\omega}{\Delta t} = \frac{\bar{\omega}_2 - \bar{\omega}_1}{t_2 - t_1} \text{ rad/s}^2$$

② Instantaneous α :

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

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* For Uniform angular acceleration, $\alpha_{avg} = \alpha_{inst}$

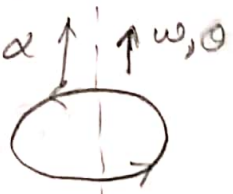
* For Uniform circular motion, $\alpha = 0$ ($\because \omega = const$)

* direction of α is along axis of rotation.

* If ω is decreasing with time.



* If ω is increasing with time.



Kinematics of circular motion:

(α — const)

* $\omega = \omega_0 + \alpha t$

* $\omega^2 - \omega_0^2 = 2\alpha\theta$

* $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

* $\theta_n = \omega_0 + \frac{\alpha}{2}(2n-1)$

* $\theta = 2\pi N$

N = no. of revolutions.

* $\theta = \left(\frac{\omega + \omega_0}{2}\right) t$.

* α — variable — $\alpha = f(\theta)$, $\alpha = f(t)$, $\alpha = f(\omega)$ } we use calculas

* $\theta \xrightarrow{\text{diff}} \omega \xrightarrow{\text{diff}} \alpha$; $\theta \xleftarrow{\text{int}} \omega \xleftarrow{\text{int}} \alpha$

$\omega = \frac{d\theta}{dt}$, $\int d\theta = \int \omega \cdot dt$

$\alpha = \frac{d\omega}{dt}$, $\int d\omega = \int \alpha \cdot dt$

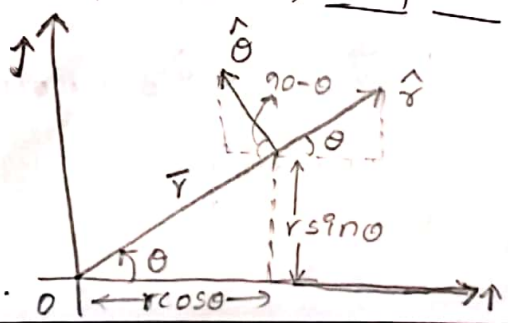
$\alpha = \omega \frac{d\omega}{d\theta} \rightarrow \int \omega \cdot d\omega = \int \alpha \cdot d\theta$

Velocity and acceleration of the particle in polar co-ordinates:

* (r, θ) → polar co-ordinates.

$\hat{r}$ = unit vector along $\vec{r}$.

$\hat{\theta}$ = unit vector ⊥ to $\vec{r}$.



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From figure,

*position vector of particle,

$$\bar{r} = r\cos\theta\,\hat{i} + r\sin\theta\,\hat{j}$$

*Velocity of the particle,

$$\bar{v} = \frac{d\bar{r}}{dt} = \frac{d}{dt}(r\cos\theta\,\hat{i} + r\sin\theta\,\hat{j})$$

$$\bar{v} = \frac{d}{dt}(r\cos\theta)\,\hat{i} + \frac{d}{dt}(r\sin\theta)\,\hat{j}$$

$$\bar{v} = r(-\sin\theta)\dot{\theta}\,\hat{i} + \dot{r}\cos\theta\,\hat{i} + r(\cos\theta)\dot{\theta}\,\hat{j} + \dot{r}\sin\theta\,\hat{j}$$

$$= \dot{r}(\cos\theta\,\hat{i} + \sin\theta\,\hat{j}) + r\dot{\theta}(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j})$$

$|\hat{r}| = 1$, $|\hat{\theta}| = 1$

$$\hat{r} = 1\times\cos\theta\,\hat{i} + 1\times\sin\theta\,\hat{j}$$

$$\boxed{\hat{r} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}}$$

$$\hat{\theta} = 1\times\cos(90-\theta)\,\hat{i} + 1\times\sin(90-\theta)\,\hat{j}$$

$$\boxed{\hat{\theta} = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}}$$

$$\bar{v} = \dot{r}(\hat{r}) + r\dot{\theta}\hat{\theta}$$

$$\boxed{\bar{v} = \frac{dr}{dt}\hat{r} + r\omega\hat{\theta}} \quad -(1) \qquad \left[\dot{r} = \frac{dr}{dt},\ \dot{\theta} = \frac{d\theta}{dt} = \omega\right]$$

$r = |\bar{r}|$

* $\bar{v}_r = \frac{dr}{dt}(\hat{r})$ = rate of change of distance along $\hat{r}$ = radial component of velocity.

* $\bar{v}_\theta = r\omega\hat{\theta}$ = velocity of $\perp$ler to $\hat{r}(\bar{r})$ = transverse velocity.

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* acceleration of the particle,

$$\bar{a} = \frac{d\bar{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{r} + r\dot{\theta}\hat{\theta})$$

$$\bar{a} = \ddot{r}\hat{r} + \dot{r}\dot{\hat{r}} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\dot{\hat{\theta}} \rightarrow (2)$$

* $\hat{r} = \cos\theta\uparrow + \sin\theta\rightarrow$

$$\dot{\hat{r}} = -(\sin\theta)\dot{\theta}\uparrow + (\cos\theta)\dot{\theta}\rightarrow$$

$$= \dot{\theta}(-\sin\theta\uparrow + \cos\theta\rightarrow)$$

$$\boxed{\dot{\hat{r}} = \dot{\theta}\times\dot{\theta}}$$

& $\hat{\theta} = -\sin\theta\uparrow + \cos\theta\rightarrow$

$$\dot{\hat{\theta}} = (-\cos\theta)\dot{\theta}\uparrow - (\sin\theta)\dot{\theta}\rightarrow$$

$$= -\dot{\theta}(\cos\theta\uparrow + \sin\theta\rightarrow)$$

$$\boxed{\dot{\hat{\theta}} = -\dot{\theta}\hat{r}}$$

eqn (2) becomes,

$$\bar{a} = \ddot{r}\hat{r} + \dot{r}\dot{\hat{r}} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}\dot{\hat{\theta}}$$

$$\bar{a} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\cdot\dot{\theta} + \dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta} + r\dot{\theta}(-\dot{\theta}\hat{r})$$

$$= \ddot{r}\hat{r} - r(\dot{\theta})^2\hat{r} + 2\dot{r}\dot{\theta}\hat{\theta} + r\ddot{\theta}\hat{\theta}$$

$$= \left[\ddot{r} - r(\dot{\theta})^2\right]\hat{r} + \left[2\frac{dr}{dt}\left(\frac{d\theta}{dt}\right) + r\frac{d^2\theta}{dt^2}\right]\hat{\theta}$$

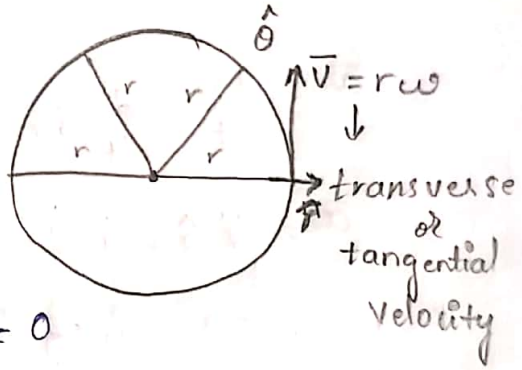
* $\bar{a}_r = \left(\frac{d^2r}{dt^2} - r\omega^2\right)\hat{r}$ = radial component acceleration of particle.

$\bar{a}_\theta = \left(2\omega\cdot\frac{dr}{dt} + r\alpha\right)\hat{\theta}$ = transverse component of acceleration.

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* Imp points

(1) $\bar{v} = \frac{dr}{dt}\hat{r} + r\omega\hat{\theta}$



In circular motion,

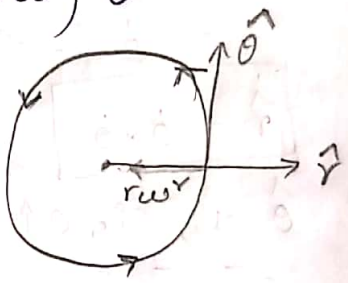
$\frac{dr}{dt} = 0$ & $\frac{d^2r}{dt^2} = 0$

(2) $\bar{a} = \left(\frac{d^2r}{dt^2} - r\omega^2\right)\hat{r} + \left(2\omega\frac{dr}{dt} + r\omega\right)\hat{\theta}$

In circular motion,

$\boxed{\bar{a} = -r\omega^2\hat{r} + r\alpha\hat{\theta}}$

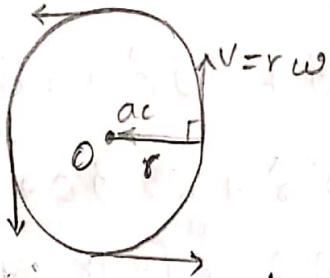
↳ linear acceleration.



$\bar{a}_c = -r\omega^2\hat{r}$ — radial component of normal comp of radial acceleration of centripetal acceleration.

$\bar{a}_t = r\alpha\hat{\theta}$ — transverse component of tangential component of acceleration. (⊥ler to $\bar{r}$, along $\bar{v}$)

(3) $\boxed{\bar{a} = \bar{a}_c + \bar{a}_t}$



$a_c = r\omega^2$

$= v\omega$

$= \frac{v^2}{r}$

clearly, $\bar{a}_c \perp \bar{a}_t$

Uniform circular motion-

$v = \text{const}$

$\omega = \text{const}$

$\alpha = 0$

Here, $a_t = 0$

$\boxed{a = a_c}$

Here angle b/w $\bar{a}$ & $\bar{v}$ is 90°.

where,

→ $\bar{r}$ = position vector

→ $\bar{v} = \frac{d\bar{r}}{dt}$ = velocity of particle

$v = |\bar{v}|$ = speed of particle.

→ $\bar{a} = \frac{d\bar{v}}{dt}$ = net acceleration of particle.

→ $a_t = \frac{d|\bar{v}|}{dt}$ = tangential acceleration

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$$\bar{a} = \bar{a_c} + \bar{a_t}$$

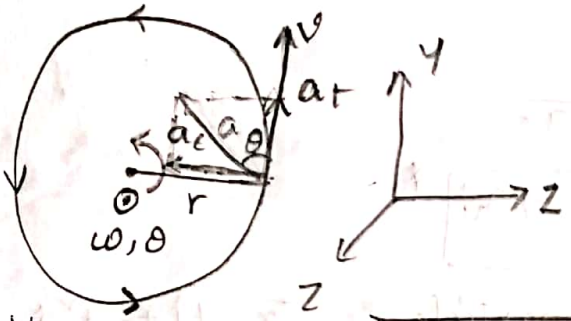
$$a = \sqrt{a_c^2 + a_t^2}$$

& $a_c^2 = a^2 - a_t^2 \Rightarrow \boxed{a_c = \sqrt{a^2 - a_t^2}}$

Non Uniform Circular Motion:

(1) Accelerated Motion

$\frac{dv}{dt} = +ve$ i.e, $v \uparrow$ & $\omega \uparrow$



* Net acceleration,

$a_c(-\uparrow) + a_t(\uparrow) + \alpha(\hat{k})$

linear components of acceleration..

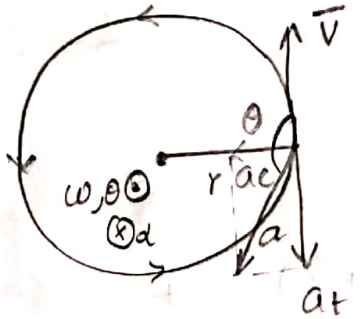
$\bar{a} \cdot \bar{v} = +ve$

$\boxed{\bar{a} = -a_c\uparrow + a_t\uparrow + \alpha\hat{k}}$

Here, $\theta < 90°$

(2) Decelerated Motion

$\frac{dv}{dt} = -ve \Rightarrow \bar{v} \downarrow$ & $\omega(\downarrow)$



* Net acceleration,

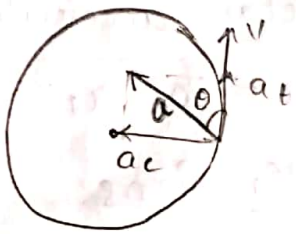
$\bar{a} = a_c(-\uparrow) + a_t(-\uparrow) + \alpha(-\hat{k})$

$\boxed{\bar{a} = -a_c\uparrow - a_t\uparrow - \alpha\hat{k}}$

$\boxed{\bar{a} \cdot \bar{v} = -ve}$

Here, $\theta > 90°$

** From figure,



$a_t = a\cos\theta$

$a_c = a\sin\theta$

$\Big\} \tan\theta = \frac{a_c}{a_t}$

or $\theta = \tan^{-1}(a_c/a_t)$ with $\bar{v}$.

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we have,

$\bar{a} \cdot \bar{v} = av\cos\theta$

$a_t = a\cos\theta = \frac{\bar{a}\cdot\bar{v}}{v} = \bar{a}\cdot\hat{v}$

$$\boxed{a_t = \bar{a}\cdot\hat{v}}$$

$$\therefore \boxed{\bar{a_t} = (a_t)\hat{v} = (\bar{a}\cdot\hat{v})\hat{v}}$$

we have

$\bar{a} = \bar{a_t} + \bar{a_c}$

$\bar{a} - \bar{a_t} = \bar{a_c}$

$\bar{a} - (\bar{a}\cdot\hat{v})\hat{v} = \bar{a_c}$

$$\boxed{\bar{a_c} = \bar{a} - (\bar{a}\cdot\hat{v})\hat{v}} \ *$$

→ position vector of a particle is $\bar{r} = t\hat{\imath} + \frac{1}{2}t^2\hat{\jmath} + t\hat{k}$

velocity of the particle,

$\bar{v} = \frac{d\bar{r}}{dt} = \hat{\imath} + t\hat{\jmath} + \hat{k}$

speed $|\bar{v}| = \sqrt{t^2+2}$ ✓

acceleration of particle,

$\bar{a} = \frac{d\bar{v}}{dt} = \hat{\jmath} \Rightarrow |\bar{a}| = 1$

tangential acceleration,

$a_t = \bar{a}\cdot\hat{v} = \hat{\jmath}\cdot\frac{(\hat{\imath} + t\hat{\jmath} + \hat{k})}{\sqrt{t^2+2}} = \frac{t}{\sqrt{t^2+2}}$

or $a_t = \frac{d|\bar{v}|}{dt}$

centripetal acceleration,

$a_c = \sqrt{a^2 - a_t^2} = \sqrt{1 - \frac{t^2}{t^2+2}} = \sqrt{\frac{2}{t^2+2}}$

for directions:

$\bar{a_t} = a_t\cdot\hat{v} = \frac{t}{\sqrt{t^2+2}}\left(\frac{\hat{\imath} + t\hat{\jmath} + \hat{k}}{\sqrt{t^2+2}}\right) = \frac{t\hat{\imath} + t^2\hat{\jmath} + t\hat{k}}{t^2+2}$

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$$\vec{a_c} = \vec{a} - \vec{a_t}$$

$$= \hat{j} - \frac{t\hat{i} + t^2\hat{j} + t\hat{k}}{t^2+2}$$

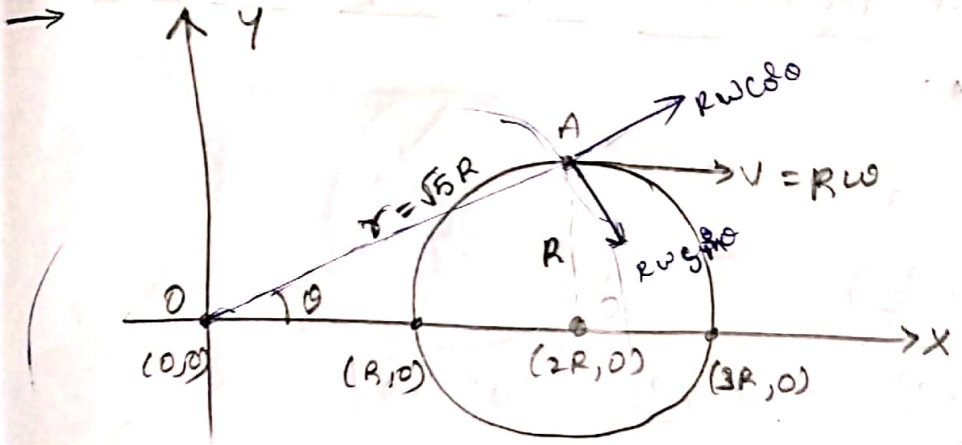
$$= \frac{t^2\hat{j} + 2\hat{j} - t\hat{i} - t^2\hat{j} - t\hat{k}}{t^2+2}$$

$$\vec{a_c} = \frac{-t\hat{i} + 2\hat{j} - t\hat{k}}{t^2+2}$$

finally,

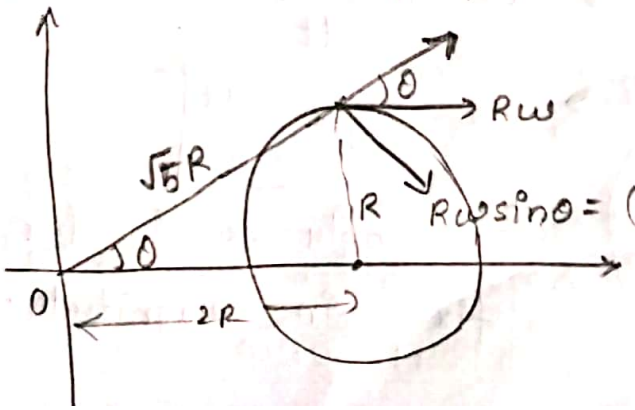
$$\vec{a} \cdot \vec{v} = \hat{j} \cdot (\hat{i} + t\hat{j} + \hat{k}) = t = +ve$$

$\therefore$ it is accelerated motion.



A particle is circulating uniform circular motion with angular velocity 'ω' on x-y plane as shown. with radius R & centre (2R,0), when it is at point A - Answer the following questions.

① Angular velocity of the particle about origin is



$$\omega_0 = \frac{(V_{AO})_\perp}{\sqrt{5}R}$$

$$= \frac{R\omega\sin\theta}{\sqrt{5}R}$$

$$= \frac{\omega}{\sqrt{5}} \times \frac{R}{\sqrt{5}R}$$

$$\boxed{\omega_0 = \omega/5}$$

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② Radial component of velocity w.r.to orgin.

$$V_T = R\omega\cos\theta$$

$$= R\omega\left(\frac{2R}{\sqrt{5}R}\right)$$

$$\boxed{V_T = \frac{2R\omega}{\sqrt{5}}}$$

③ Net acceleration of the particle w.r.to origin.

* since particle is in uniform circular motion.

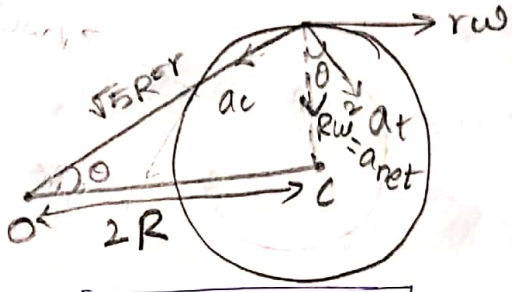
$a_{net} = R\omega^2$, towards centre of circle.

④ Transverse acceleration of the particle w.r.to origin.

$$a_t = R\omega^2\cos\theta$$

$$= R\omega^2\left(\frac{2}{\sqrt{5}}\right)$$

$$\boxed{a_t = \frac{2}{\sqrt{5}}R\omega^2}$$

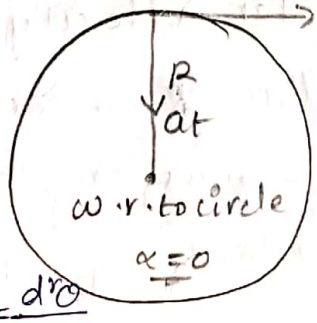


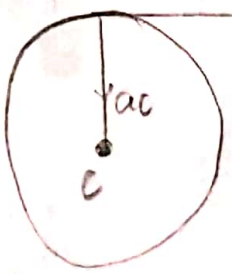
& $\boxed{a_c = R\omega^2\sin\theta}$

⑤ Magnitude of angular acceleration of the particle w.r.to origin.

$a_t = R\alpha$ is applicable for R = const

Otherwise we use $\alpha = \frac{d^2\theta}{dt}$





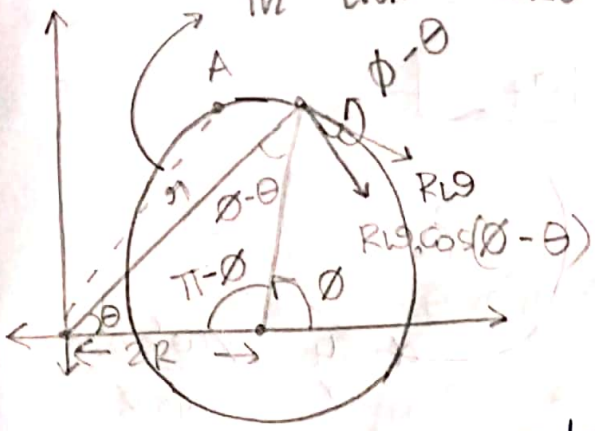
$$\boxed{\alpha = \frac{a_t}{R}}$$

is only valid for 'R' is fixed in magnitude.

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In general,

$$\boxed{\alpha = \frac{d^2\theta}{dt^2}} \rightarrow \text{valid for all the cases.}$$



In this case r changes

$$\omega = -\frac{d\phi}{dt}$$

$$\& \quad \omega_0 = \frac{-d\theta}{dt} = \frac{\omega}{5}$$

$$\omega_0 = \frac{R\omega\cos(\phi-\theta)}{r}$$

$$\frac{-d\theta}{dt} = \frac{R\omega\cos(\phi-\theta)}{r} \quad -①$$

From sine rule,

$$\frac{R}{\sin\theta} = \frac{r}{\sin(\pi-\phi)}$$

$$\frac{R}{\sin\theta} = \frac{r}{\sin\phi} \Rightarrow r = \frac{R\sin\phi}{\sin\theta}$$

$$① \Rightarrow \frac{-d\theta}{dt} = \frac{R\omega\cos(\phi-\theta)}{\frac{R\sin\phi}{\sin\theta}}$$

$$\frac{-d\theta}{dt} = \frac{\omega}{2\sin\phi}\left[2\sin\theta\cos(\phi-\theta)\right]$$

$$= \frac{\omega}{2\sin\phi}\left[\sin(\phi+\phi-\theta) + \sin(\phi-\phi+\theta)\right]$$

$$= \frac{\omega}{2\sin\phi}\left[\sin(\theta+\phi-\theta) + \sin(\theta-\phi+\theta)\right]$$

$$\frac{-d\theta}{dt} = \frac{\omega}{2\sin\phi}\left[\sin\phi + \sin(2\theta-\phi)\right]$$

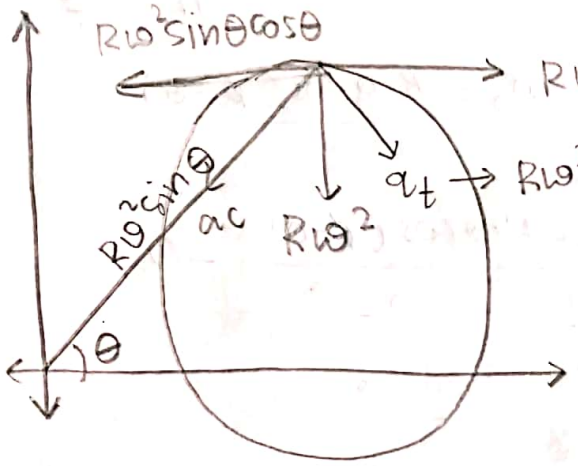
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$$\omega_0 = -\frac{d\theta}{dt} = \frac{\omega}{2}\left[1 + \frac{\sin(2\theta - \phi)}{\sin\phi}\right]$$

angular acceleration,

$$\alpha = \frac{d\omega_0}{dt} = -\frac{d^2\theta}{dt^2} = \frac{d}{dt}\left\{\frac{\omega}{2}\left[1 + \frac{\sin(2\theta-\phi)}{\sin\phi}\right]\right\}$$

$$\boxed{\alpha = \frac{6\omega^2}{25}} \qquad \boxed{a_t = 0}$$



$$\alpha = \frac{a_t}{r} = \frac{R\omega^2\cos\theta}{\sqrt{5}R}$$

$$= \frac{\omega^2}{\sqrt{5}} \times \frac{2}{\sqrt{5}} = \frac{2\omega^2}{\sqrt{5}}$$ ✓

it is not correct.

Pseudo Force :

* It is the fictitious force acting on a body when placed in accelerating / rotating frame.
* Centrifugal force is one of the components of pseudo force acting on a body in rotating frame.

$$\boxed{F_c = mr\omega^2}$$ *

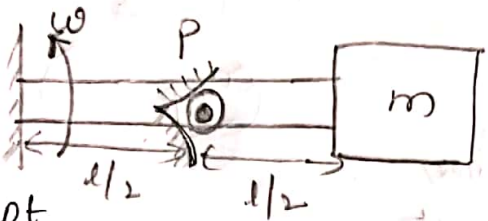
→ A block of mass m is attached to a rigid rod of length l. moving in circular path with angular velocity about point 'O'. observes is fixed.

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Find: ① Velocity of block 'm'.

② Acceleration of block 'm'.

③ pseudo force acting on block 'm'.

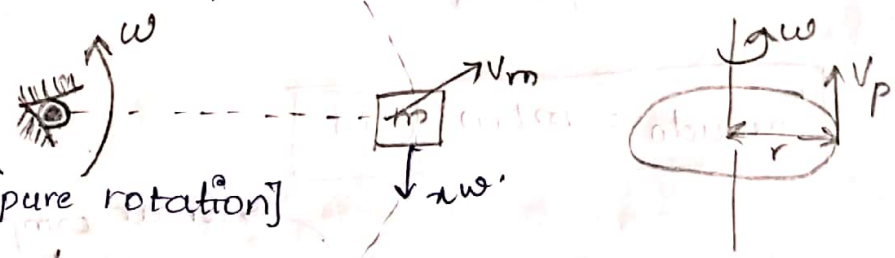


* Concept

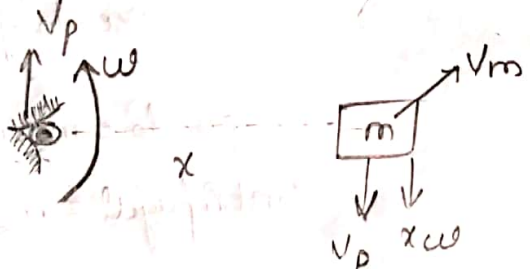
① 

$$V_{m,p} = \bar{V}_m - \bar{V}_p = \bar{V}_m + (-\bar{V}_p)$$

[pure tangential]

② 

[pure rotation]

③ 

$$\bar{V}_{mp} = \bar{V}_m - V_p + (\bar{\omega} \times \bar{x}_{pm})$$

or

$$\bar{V}_{mp} = \bar{V}_m - \bar{V}_p - (\bar{\omega} \times \bar{x}_{mp})$$

Conclusions: ① In translatory ~~(motion)~~ frame,

$$\boxed{\bar{V}_{AB} = \bar{V}_A - \bar{V}_B}$$

② In rotating frame,

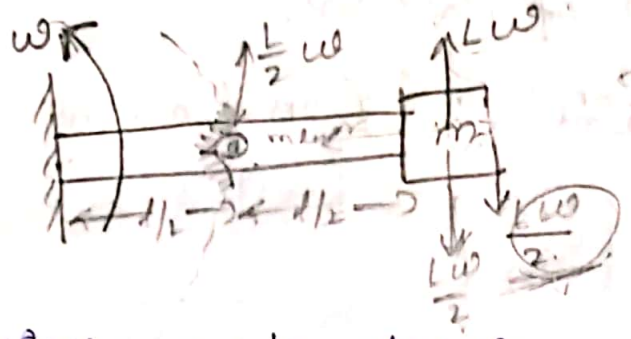
$$\boxed{\bar{V}_{AB} = \bar{V}_A - \bar{V}_B + (\bar{\omega} \times \bar{r}_{BA})}$$

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Answer to the question:

$v = r\omega$

(1)



$V_{mp} = 0$

$a_{mp} \geq 0$

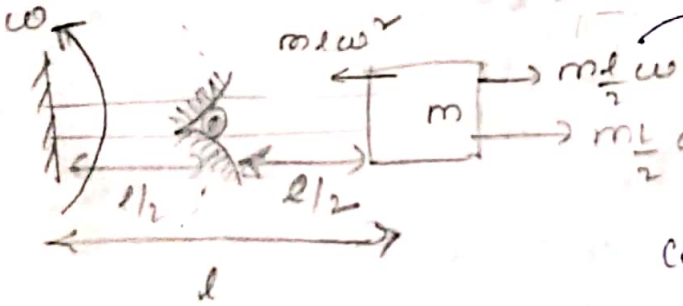
(2) since w.r.to observer, block is at rest.

$$F_{net} = 0$$

$$F_{real} + F_{pseudo} = 0$$

$$mL\omega^2(-\uparrow) + F_{pseudo} = 0$$

$$F_{pseudo} = mL\omega^2(\uparrow)$$

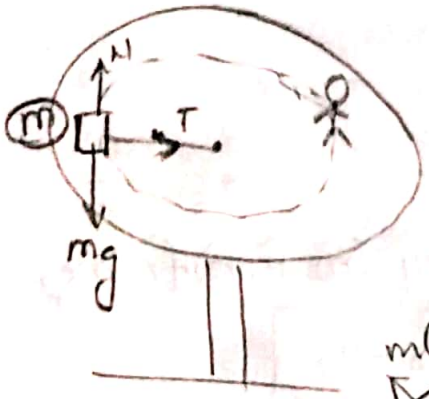


linear component of pseudo force

pseudo = $m\ell\omega^2$

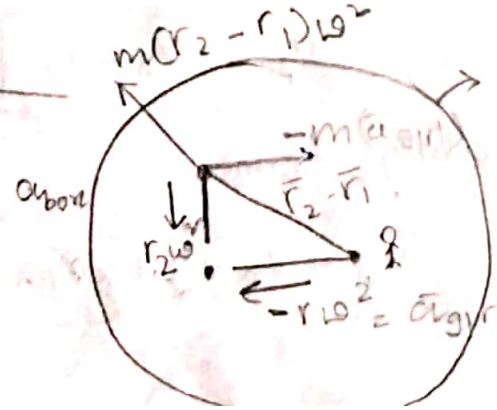
centrifugal = $m\frac{\ell}{2}\omega^2$.

→



If block is always at rest w.r.t girl. then $(\bar{F}_{box}) = 0$

$\Rightarrow \bar{F}_{real} + \bar{F}_{pseudo} = 0$



$\bar{r}_1 + \bar{x} = \bar{r}_2$

$\bar{x} = (\bar{r}_2 - \bar{r}_1)$

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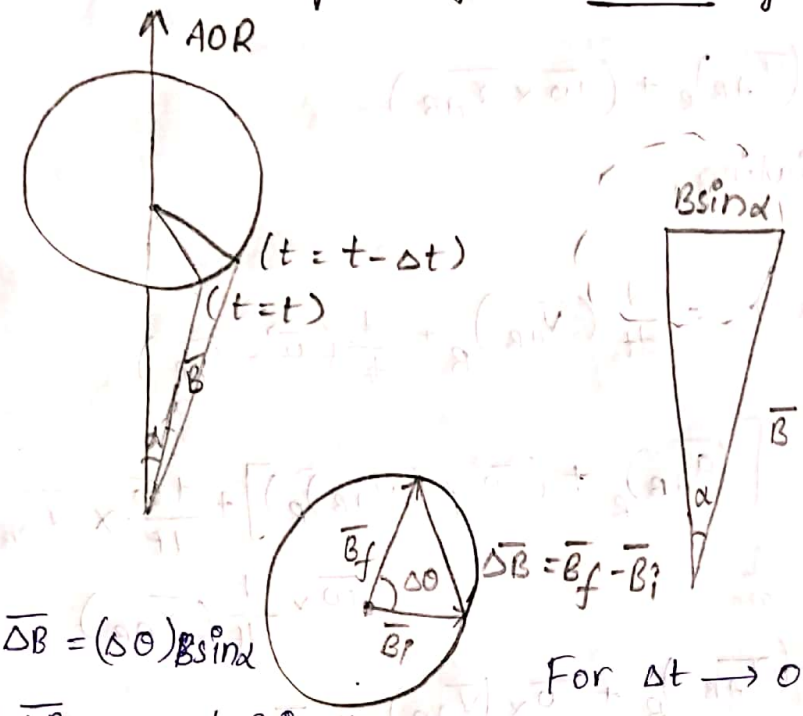
$$\bar{F}_{net} = \bar{F}_{real} + \bar{F}_{pseudo}$$

$$= m\bar{a}_{box} + m\bar{r}_1\omega^2 + m(\bar{r}_2 - \bar{r}_1)\omega^2$$

$$= -m r_2\omega^2 + m\bar{r}_1\omega^2 + m\bar{r}_2\omega^2 - m\bar{r}_1\omega^2$$

$$\boxed{\bar{F}_{net} = 0}$$

Rate of change of Rotating Vectors



$$\Delta\bar{B} = (\Delta\theta)B\sin\alpha$$

$$\Delta\bar{B} = \omega\,\Delta t\,B\sin\alpha$$

$$\boxed{\frac{\Delta\bar{B}}{\Delta t} = B\omega\sin\alpha}$$

For $\Delta t \to 0$

$$\frac{d\bar{B}}{dt} = B\omega\sin\alpha$$

In vector form,

$$\boxed{\frac{d\bar{B}}{dt} = \bar{\omega} \times \bar{B}}$$

Vector $\bar{B}$ is fixed in magnitude.

In general, $\dfrac{d\bar{A}}{dt} = \bar{\omega} \times \bar{A}$



$$\frac{d\bar{r}}{dt} = \bar{\omega} \times \bar{r}$$

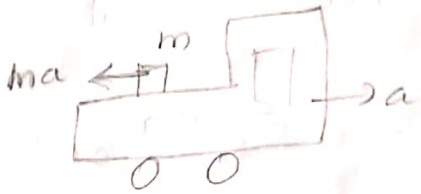
$$\boxed{\bar{v} = \bar{\omega} \times \bar{r}}$$

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Coriolis force, Euler force:

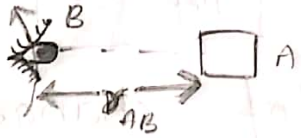
* In translating frame,

$$\bar{V}_{AB} = \bar{V}_A - \bar{V}_B$$



* In rotating frame,

$$\left(\bar{V}_{AB}\right)_R = \left(\bar{V}_A - \bar{V}_B\right) - \left(\bar{\omega} \times \bar{r}_{AB}\right)$$



$$\Rightarrow \left(\bar{V}_{AB}\right)_T = \left(\bar{V}_{AB}\right)_R + \left(\bar{\omega} \times \bar{r}_{AB}\right)$$

$$\Rightarrow \frac{d}{dt}\bar{r}_{AB} = \left(\bar{V}_{AB}\right)_R + \left(\bar{\omega} \times \bar{r}_{AB}\right)$$

on differentiating,

$$\frac{d^2}{dt^2}\left(\bar{r}_{AB}\right) = \frac{d}{dt}\left(\bar{V}_{AB}\right)_R + \frac{d}{dt}\left(\vec{\omega} \times \bar{r}_{AB}\right)$$

$$\Rightarrow \vec{a}_{AB} = \left[\left(\bar{a}_{AB}\right)_R + \left(\bar{\omega} \times \left(\bar{V}_{AB}\right)_R\right)\right] + \frac{d\bar{\omega}}{dt} \times \bar{r}_{AB} + \bar{\omega} \times \frac{d}{dt}\left(\bar{r}_{AB}\right)$$

$$\Rightarrow \bar{a}_{AB} = \left(\bar{a}_{AB}\right)_R + \bar{\omega} \times \left(\bar{V}_{AB}\right)_R + \bar{\alpha} \times \bar{r}_{AB} + \bar{\omega} \times \left[\left(\bar{V}_{AB}\right)_R + \left(\bar{\omega} \times \bar{r}_{AB}\right)\right]$$

$$\Rightarrow \bar{a}_{AB} = \left(\bar{a}_{AB}\right)_R + 2\bar{\omega} \times \left(\bar{V}_{AB}\right)_R + \bar{\omega} \times \left(\bar{\omega} \times \bar{r}_{AB}\right) + \bar{\alpha} \times \bar{r}_{AB}$$

$$\Rightarrow \left(\bar{a}_{AB}\right)_R = \bar{a}_A - \bar{a}_B - 2\bar{\omega} \times \left(\bar{V}_{AB}\right)_R - \bar{\omega} \times \left(\bar{\omega} \times \bar{r}_{AB}\right) - \bar{\alpha} \times \bar{r}_{AB}$$

$$m\left(\bar{a}_{AB}\right)_R = m\bar{a}_A - m\bar{a}_B - 2m\left(\bar{\omega} \times \left(\bar{V}_{AB}\right)_R\right) - m\bar{\omega} \times \left(\bar{\omega} \times \bar{r}_{AB}\right) - m\left(\bar{\alpha} \times \bar{r}_{AB}\right)$$

$$F_{net} = \sum \bar{F}_{real} + \bar{F}_{pseudo}$$

acting on A as seen by B.

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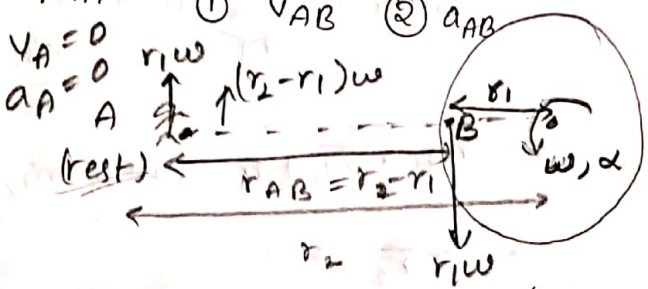
* $-m\bar{a}_B$ = linear component of pseudo force.

$-2m(\bar{\omega} \times (\bar{V}_{AB})_R)$ = coriolis force. It depends on V_{rel} but not on its position r_{AB}.

* $-m\bar{\omega} \times (\bar{\omega} \times \bar{r}_{AB})$ = centrifugal force. It depends on position r_{AB} but not V_{rel}.

* $-m(\alpha \times \bar{r}_{AB})$ = Euler force.

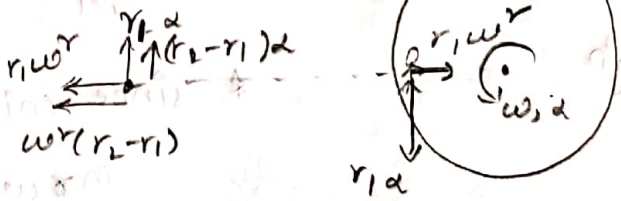
→ Find ① $\bar{V}_{AB}$ ② $\bar{a}_{AB}$



$$(\bar{V}_{AB})_R = (\bar{V}_A - \bar{V}_B) - (\bar{\omega} \times \bar{r}_{AB})$$

$$\bar{V}_{AB} = r_1\omega + r_2\omega - r_1\omega$$

$$\bar{V}_{AB} = r_2\omega\,(\uparrow)$$



$$-2(\bar{\omega} \times (\bar{V}_{AB})_R)$$

$$= -2\bar{\omega}(\hat{k}) \times (\bar{r}_2\omega\hat{j})$$

$$= -2\omega^2 r_2 \uparrow$$

$$-\bar{\omega} \times (\bar{\omega} \times \bar{r}_{AB})$$

$$= \omega^2(r_2 - r_1)(\hat{k} \times \hat{j})$$

$$= \omega^2(r_2 - r_1)(-\uparrow)$$

$$-(\bar{\alpha} \times \bar{r}_{AB}) = -\alpha\hat{k} \times (r_2 - r_1)(-\uparrow)$$

$$= \alpha(r_2 - r_1)(\hat{k} \times \hat{\imath})$$

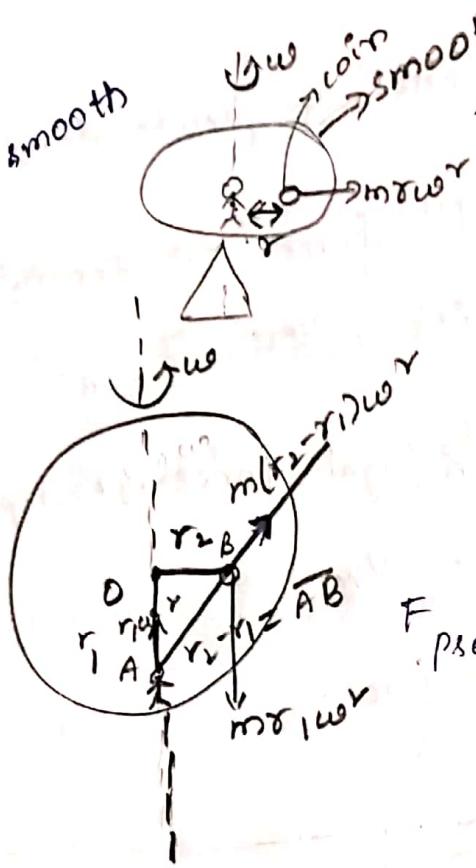
$$= \alpha(r_2 - r_1)(\hat{j})$$

$$\bar{a}_{AB} = r_2\omega^2(\uparrow) + \alpha(r_2)(\hat{j})$$

$$= r_2(\omega^2\uparrow + \alpha\hat{j})$$

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→



$= -m\vec{\omega}\times(\vec{\omega}\times\vec{r})$

$= -m\omega\hat{j}\,(\omega\hat{j}\times r\hat{k})$

$= -mr\omega^2(\hat{j}\times\hat{k})$

$= -mr\omega^2(-\hat{i})$

$= mr\omega^2\hat{i}$

$F_{pseudo} = mr_1\omega^2(\hat{OA}) + m(r_2-r_1)\omega^2(\hat{AB})$

$= m r_1\omega^2\frac{\vec{r_1}}{r_1} + m(r_2-r_1)\omega^2\frac{(\vec{r_2}-\vec{r_1})}{r_2-r_1}$

$= mr_1\omega^2\vec{r_1} + mr_2\omega^2 - mr_1\omega^2$

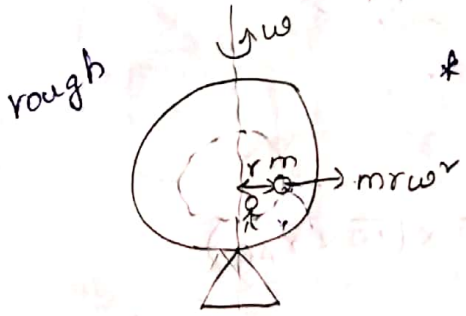
$= mr_2\omega^2$

when coin starts slipping on table

Coriolis force $= -2m(\vec{\omega}\times\vec{V}_{rel})$

$= 2mr\omega v_{rel}$

→



* since coin is a rest,

$mr\omega^2 = f$

$f \le f_s$

$mr\omega^2 \le \mu_s N$

$mr\omega^2 \le \mu_s mg$

$r\omega^2 \le \mu_s g$

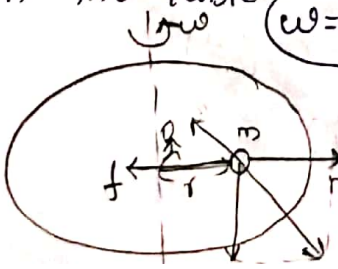
$\omega^2 \le \frac{\mu_s g}{r}$

$\boxed{\omega \le \sqrt{\frac{\mu_s g}{r}}}$

* $T_{min} = \frac{2\pi}{\omega_{max}} = 2\pi\sqrt{\frac{r}{\mu g}}$

* If $\omega > \omega_{max}$, coin slips on the table

$\omega = \omega_0 + \alpha t$



* Coriolis force $= -2m(\vec{\omega}\times\vec{V}_{rel})$

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$= -m(\alpha\hat{\uparrow} \times r\hat{\uparrow})$

$= m\alpha r\hat{k}$

$F_{net} = \sqrt{(mr\omega^2)^2 + (mr\alpha)^2}$

* coin is at rest on table,

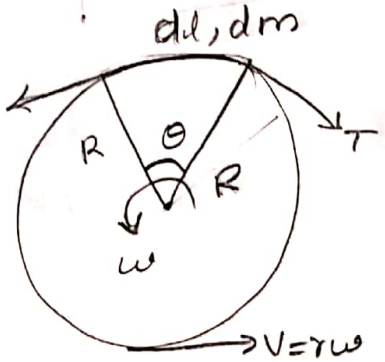
$F = f$ (static friction)

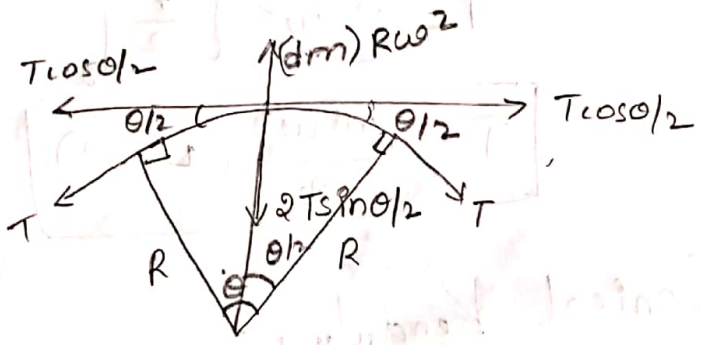
In ground,

$f \leq f_s$

$$\boxed{\sqrt{(mr\omega^2)^2 + (mr\alpha)^2} \leq \mu_s mg}$$

Tension in the chain / ring (in rotating frame):





$2T\sin\theta/2 = (dm)R\omega^2$

$2T\sin\theta/2 = \left(\frac{m}{L}dl\right)R\omega^2$

$T\left(\frac{dl}{R}\right) = \frac{m}{L}dl\,R\omega^2$

$T = \frac{mR^2\omega^2}{L}$

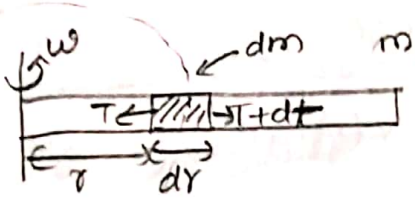
$T = \frac{mR^2\omega^2}{2\pi R} = \frac{mR\omega^2}{2\pi}$

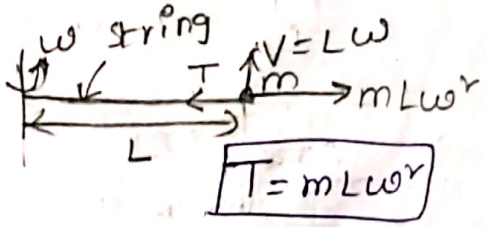
$$\boxed{T = \frac{mR\omega^2}{2\pi}}$$

$$T = \frac{mv^2}{L}$$

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Tension at a point in whirling rope :-





$T = mL\omega^2$

$T - (T+dT) = (dm)\, r\omega^2$

$$-\int_T^0 dT = \int_r^L \frac{m}{l}\, dr \cdot r\omega^2$$

$$T = \frac{m}{L}\omega^2 \int_r^L r\, dr$$

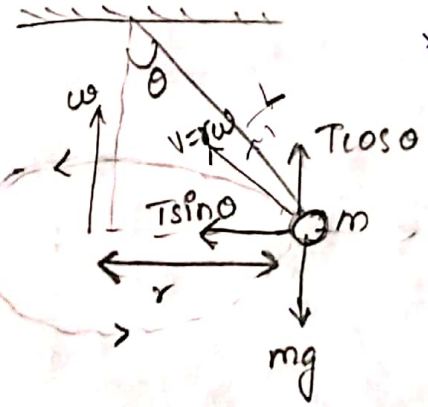
on AOR, $r = 0$

$$T = \frac{m}{L}\omega^2 \left[\frac{r^2}{2}\right]_r^L$$

$$\boxed{T = \frac{m\omega^2}{2L}(L^2 - r^2)}$$

$$\boxed{T_{max} = \frac{mL\omega^2}{2}}$$

Conical Pendulum :



* $\sin\theta = \frac{l}{L} \Rightarrow l = L\sin\theta$

$\cos\theta = \frac{\sqrt{L^2 - l^2}}{L}$

* $T\cos\theta = mg$

$T\sin\theta = mr\omega^2$

$T\sin\theta = m(L\sin\theta)\omega^2$

$\boxed{T = mL\omega^2}$

$\therefore mL\omega^2\cos\theta = mg$

*As $\omega \uparrow \Rightarrow \cos\theta$ ($\downarrow$)

$\Rightarrow \theta(\uparrow)$

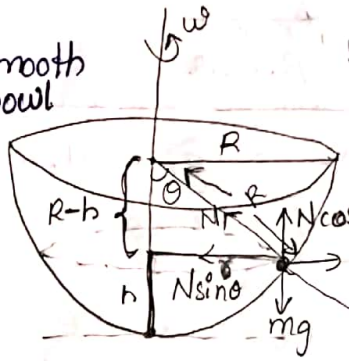
$$\omega = \sqrt{\frac{g}{L\cos\theta}}$$

* Time period, $T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$

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Particle in a Rotating bowl:

*Smooth bowl



$\sin\theta = \frac{r}{R}$ $\quad N\cos\theta = mg$

$r = R\sin\theta$ $\quad N\sin\theta = mr\omega^2$

$\tan\theta = \frac{r\omega^2}{g}$

$\frac{\sin\theta}{\cos\theta} = \frac{R\sin\theta\,\omega^2}{g}$

$\boxed{\cos\theta = \frac{g}{R\omega^2}}$

* $\sin\theta = \frac{r}{R} \Rightarrow \boxed{r = R\sin\theta}$

* $\omega^2 = \frac{g}{R\cos\theta} \Rightarrow \boxed{\omega = \sqrt{\frac{g}{R\cos\theta}}}$

* $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{R\cos\theta}{g}}$

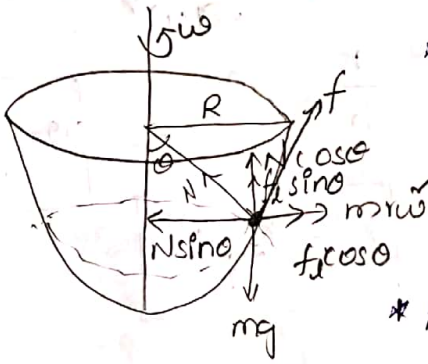
* Again, $\cos\theta = \frac{R-h}{R}$

* $\omega^2 = \frac{g}{R-h} \Rightarrow \boxed{\omega = \sqrt{\frac{g}{R-h}}}$

$\frac{g}{R\omega^2} = \frac{R-h}{R}$

$\boxed{h = R - \frac{g}{\omega^2}}$

* rough bowl



* If $\omega < \sqrt{\frac{g}{R\cos\theta}}$

* For $\omega = \omega_{min}$, $f_{max} = f = \mu_s N$

* $N\cos\theta + f_s\sin\theta = mg$

$N\cos\theta + \mu_s N\sin\theta = mg$

* $N\sin\theta - f_s\cos\theta = mr\omega^2$

$N = \frac{mg}{\cos\theta + \mu_s\sin\theta}$ $\quad -①$

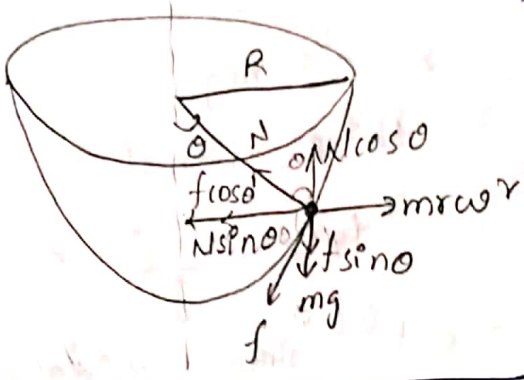
$N\sin\theta - \mu_s N\cos\theta = mr\omega^2$

$N = \frac{mr\omega^2}{\sin\theta - \mu_s\cos\theta}$ $\quad -②$

$\frac{①}{②} \Rightarrow \frac{mr\omega_{min}^2}{mg} = \left(\frac{\cos\theta + \mu\sin\theta}{\sin\theta - \mu_s\cos\theta}\right)^{-1} \Rightarrow \boxed{\omega_{min} = \sqrt{\frac{g}{r}\left(\frac{\sin\theta - \mu_s\cos\theta}{\cos\theta + \mu\sin\theta}\right)}}$

min $\quad \frac{r\omega^2}{g} = \tan(\theta - \lambda)$

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* $\omega > \sqrt{\frac{g}{R\cos\theta}}$

$$\omega_{max} = \sqrt{\frac{g}{r}\left(\frac{\sin\theta + \mu_s\cos\theta}{\cos\theta - \mu_s\sin\theta}\right)}$$

$$\tan(\theta+\lambda) = \frac{r\omega^2}{g}$$

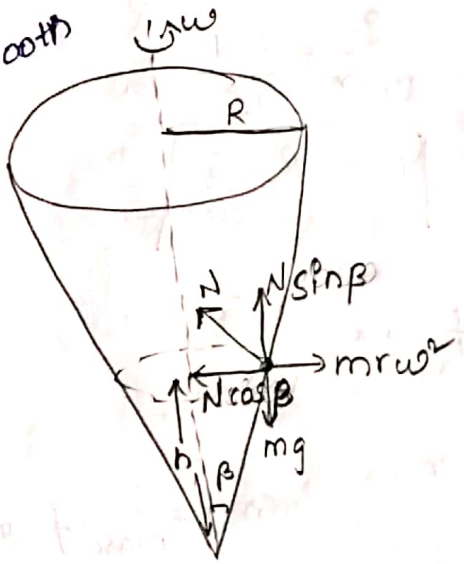
↳ max

For $\omega_{min} \le \omega \le \omega_{max}$

particle is at rest w.r.to bowl at same height

Particle in a Rotating cone:

*smooth



$N\sin\beta = mg$

$N\cos\beta = mr\omega^2$

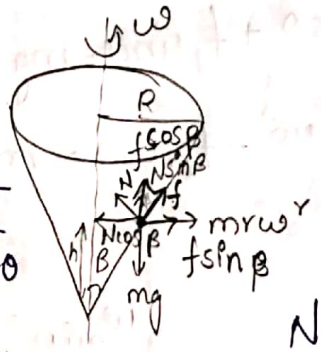
$$\tan\beta = \frac{g}{r\omega^2} = \frac{gr}{v^2}$$

$$\frac{r}{h} = \frac{g}{r\omega^2}$$

$$\omega = \sqrt{\frac{g}{r\tan\theta}}$$

$$h = \frac{r^2\omega^2}{g}$$

*rough



$\omega \ge \sqrt{\frac{g}{r\tan\theta}}$

$\omega_{max} = \omega$

$f_{max} = f = \mu_s N$

* $N\sin\beta + f\cos\beta = mg$

* $N\cos\beta = mr\omega^2 + f\sin\beta$

$N\sin\beta + \mu_s N\cos\beta = mg$

$N\cos\beta - \mu_s N\sin\beta = mr\omega^2$

$$N = \frac{mg}{\sin\beta + \mu_s\cos\beta} \quad -①$$

$$N = \frac{mr\omega^2}{\cos\beta - \mu_s\sin\beta} \quad -②$$

* As v↑ then h↑

As v↓ then h↓

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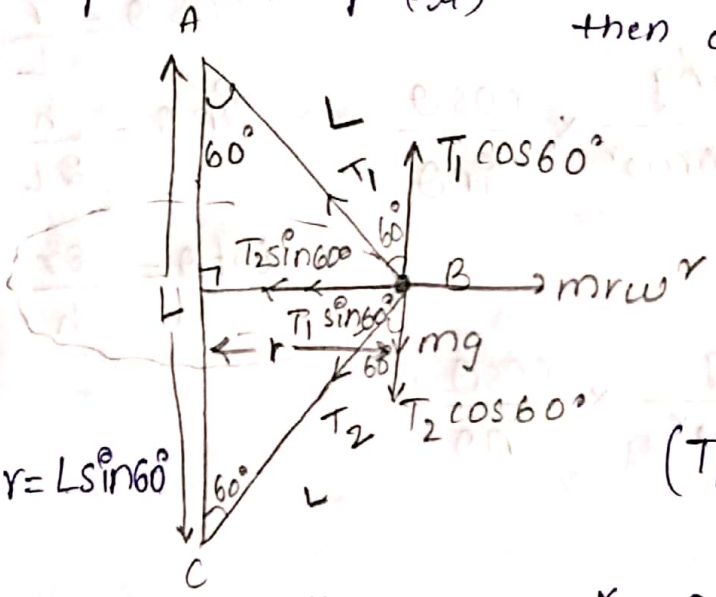
$\frac{①}{②} \Rightarrow \frac{mr\omega^2_{min}}{mg} = \frac{\cos\beta - \mu_s \sin\beta}{\sin\beta + \mu_s \cos\beta}$

$\therefore \omega^2 r = \frac{g}{r}\left(\frac{1-\mu\tan\beta}{\mu+\tan\beta}\right) \times \frac{1}{\tan\beta}$

$$\omega_{min} = \sqrt{\frac{g}{h\tan\beta}\left(\frac{1-\mu\tan\beta}{\mu+\tan\beta}\right)}$$

replace (μ) by ($-\mu$) then we get ω_{max}.

→



$T_1 \sin 60° + T_2 \sin 60° = mr\omega^2$

$(T_1 + T_2)\frac{\sqrt{3}}{2} = mr\omega^2$

$(T_1 - T_2)\cos 60° = mg$

$\frac{T_1 - T_2}{2} = mg$

$T_1 + T_2 = mw^2 L - ①$

$T_1 - T_2 = 2mg - ②$

Solve ① & ② for T_1 & T_2.

→ when tension in lower string is zero i.e, $T_2 = 0$

then $T_1 = 2mg$

$T_1 = mw^2 L$.

$mw^2 L = 2mg$

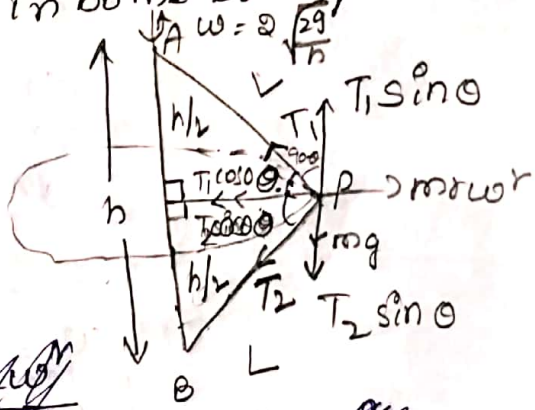
$$\boxed{\omega = \sqrt{\frac{2g}{L}}}^{*}$$

i.e, for $\omega \geq \sqrt{\frac{2g}{L}}$ T_2 becomes zero.

for $\omega > \sqrt{\frac{2g}{L}}$ then $T_2 \neq 0$.

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→ Find ratio of tensions in both the strings.



$$T_1 \sin\theta = T_2 \sin\theta + mg$$

$$T_1 - T_2 = \frac{mg}{\sin\theta}$$

$$T_1 + T_2 = \frac{mr\omega^2}{\cos\theta}$$

$$\cos\theta = \frac{r}{L}$$

$$\sin\theta = \frac{h}{2L}$$

$$\cot\theta = \frac{2r}{h}$$

$$\frac{T_1 - T_2}{T_1 + T_2} = \frac{mg}{mr\omega^2} \times \frac{\cos\theta}{\sin\theta}$$

$$\frac{\frac{T_1}{T_2} - 1}{\frac{T_1}{T_2} + 1} = \frac{g}{r\omega^2} \times \frac{\cos\theta}{\sin\theta}$$

$$\frac{t-1}{t+1} = \frac{g}{r\omega^2} \times \frac{r}{L\sin\theta} = \frac{g \times L}{\omega^2 \times r \times \frac{h}{2}}$$

$$\frac{t-1}{t+1} = \frac{2g}{\omega^2 \times h}$$

$$= \frac{2g}{4\left(\frac{2g}{h}\right)(h)} = \frac{1}{4}$$

$$4(t-1) = t+1$$

$$4t - 4 = t + 1$$

$$3t = 5$$

$$t = 5/3$$

$$\therefore \boxed{\frac{T_1}{T_2} = \frac{5}{3}}$$

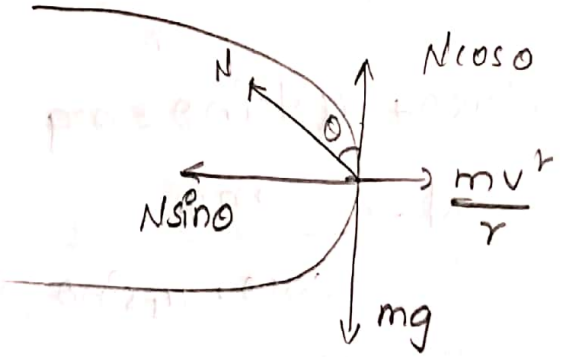
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Bending of a cycle while taking turn:

* $N\cos\theta = mg$

$N\sin\theta = \frac{mv^2}{r}$

$$\tan\theta = \frac{v^2}{rg}$$



$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$ with vertical.

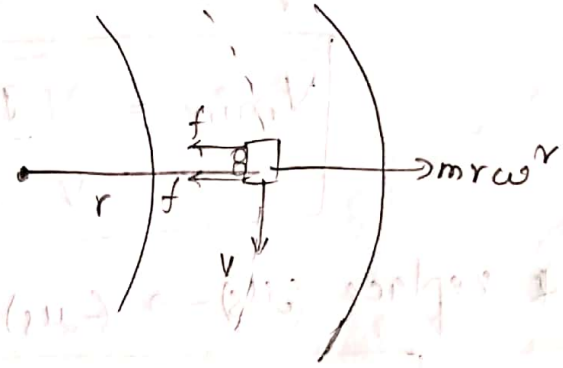
* $f = \frac{mv^2}{r}$

$f \le f_d$

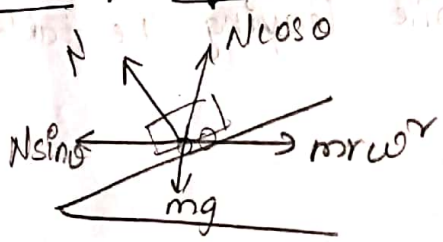
$\frac{mv^2}{r} \le \mu_s mg$

$v \le \sqrt{\mu_s rg}$

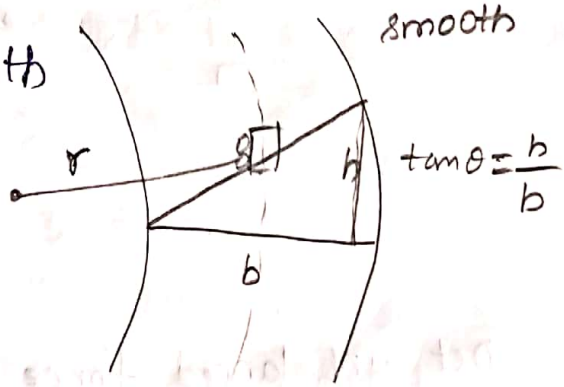
$$(v_{max})_{\text{safe speed}} = \sqrt{\mu rg}$$



Banking of roads:



Smooth



$N\cos\theta = mg$

$N\sin\theta = mr\omega^2$

$\tan\theta = \frac{v^2}{rg} \Rightarrow$ $\boxed{v = \sqrt{rg\tan\theta}}$

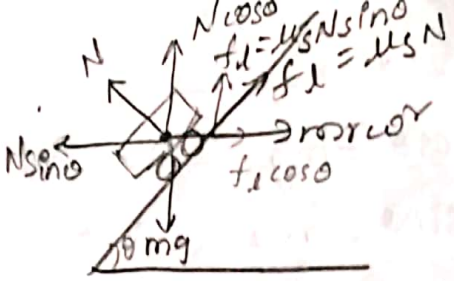
Rough banked Roads

* $v < \sqrt{gr\tan\theta}$ then,

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$N\cos\theta + f_s\sin\theta = mg$

$N\sin\theta - f_s\cos\theta = m\frac{v_{min}^2}{r}$



$N\cos\theta + \mu_s N\sin\theta = mg$

$N = \frac{mg}{\cos\theta + \mu_s\sin\theta}$

$N\sin\theta - \mu_s\cos\theta = \frac{mv_{min}^2}{r}$

$N = \frac{mv_{min}^2}{r(\sin\theta - \mu_s\cos\theta)}$

$\frac{mv_{min}^2}{mg(r)} = \frac{\sin\theta - \mu_s\cos\theta}{\cos\theta + \mu_s\sin\theta}$

$\frac{v_{min}^2}{gr} = \tan(\theta - \lambda)$; $\lambda = \tan^{-1}\mu_s$

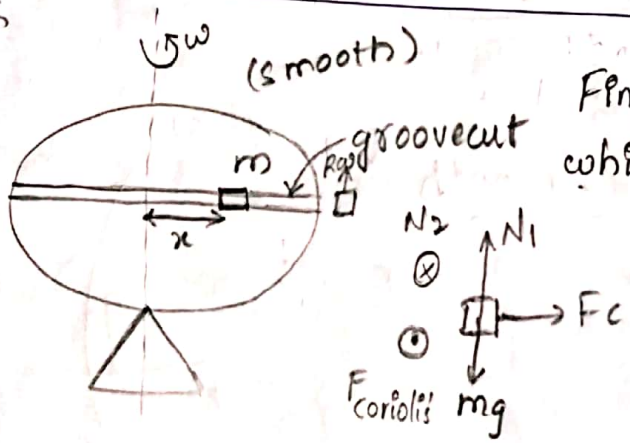
$$v_{min} = \sqrt{\frac{gr(\sin\theta - \mu_s\cos\theta)}{\cos\theta + \mu_s\sin\theta}}$$

replace $(\mu_s) \rightarrow (-\mu_s)$

$$v_{max} = \sqrt{\frac{gr(\sin\theta + \mu_s\cos\theta)}{\cos\theta - \mu_s\sin\theta}}$$

$\frac{v_{max}^2}{gr} = \tan(\theta + \lambda)$; $\lambda = \tan^{-1}\mu_s$

$v > \sqrt{gr\tan\theta}$, f acts downwards.



Find the speed of block while leaving the table.

Net unbalanced force, $F_c = mr\omega^2$

$ma = mr\omega^2$

$v\cdot\frac{dv}{dx} = x\omega^2$

$\int_0^{v_1} v\,dv = \int_x^R x\omega^2\,dx$

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$$\left[\frac{v^2}{2}\right]_0^{v_1} = \left[\frac{x^2}{2}\right]_x^R \omega^2 \cdot dx$$

$$\frac{v_1^2}{2} = \frac{R^2 - x^2}{2}(\omega^2)$$

$$\boxed{v_1 = \left(\sqrt{R^2 - x^2}\right)\omega}$$ — relative radial speed of block w.r.to table.

$v_2 = R\omega$ — tangential speed of block/table.

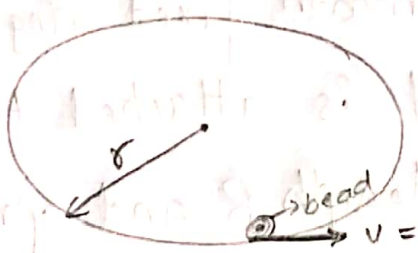
$$\bar{v} = v_1 \hat{\uparrow} + v_2(-\hat{k})$$

$$|\bar{v}| = \sqrt{v_1^2 + v_2^2}$$

$$|\bar{v}| = \sqrt{\omega^2(R^2 - x^2) + (R^2\omega^2)}$$

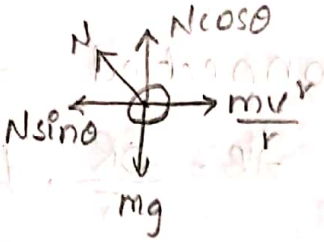
$$\boxed{|\bar{v}| = \omega\sqrt{2R^2 - x^2}}$$

→ Find distance travelled by bead before coming to rest.



v = initial velocity

μ_K = coefficient of friction b/w bead & ring.



$N\cos\theta = mg$

$N\sin\theta = \frac{mv^2}{r}$

$$N^2 = m^2\left(g^2 + \frac{v^4}{r^2}\right)$$

$$\boxed{N = m\sqrt{g^2 + \frac{v^4}{r^2}}}$$

$f_K = \mu_K N$

$$-ma = \mu m\sqrt{g^2 + \frac{v^4}{r^2}}$$

$$\frac{-v\,dv}{ds} = \frac{\mu}{r}\sqrt{g^2r^2 + v^4}$$

$$\int_{v_0}^{0} \frac{-2v\,dv}{\sqrt{g^2r^2 + v^4}} = \frac{2\mu_K}{r}\int_0^S ds$$

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$$\int_0^{v_0} \frac{dv^r}{\sqrt{g^r r^r + (v^2)^r}} = \frac{2\mu_k}{r}(s)$$

we have, $\int \frac{dx}{\sqrt{a^r + x^r}} = \ln\left[x + \sqrt{a^r + x^r}\right]$

$$\left[\ln\left[v^r + \sqrt{g^r r^r + v^4}\right]\right]_0^{v_0} = \frac{2\mu_k s}{r}$$

$$\ln\left[v_0^r + \sqrt{g^r r^r + v^4}\right] - \ln gr = \frac{2\mu_k s}{r}$$

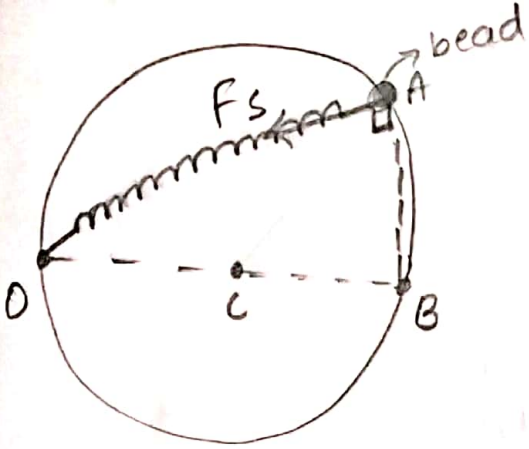
$$\boxed{s = \frac{r}{2\mu_k} \ln\left[\frac{v_0^r + \sqrt{g^r r^r + v^4}}{gr}\right]}$$

A bead of mass $m = 300$gms moves in gravity free region along a smooth fixed ring of radius $R = 2$m. The bead is attached to a spring having natural length R and spring constant $K = 10$N/m. The other end of spring is connected to a fixed point O on the ring $AB = \frac{6R}{5}$. Line OB is diameter of ring. Find a speed of bead at A if normal reaction on bead due to ring at A is zero.

$m = 300$gms,

$K = 10$N/m, $AB = \frac{6R}{5}$ $R = 2$cm.

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Elongation in spring,

$$x = OA - R$$

$$= \sqrt{(2R)^2 - \left(\frac{6R}{5}\right)^2} - R$$

$$= R\sqrt{4 - \frac{36}{25}} - R$$

$$= \frac{8R}{5} - R = \frac{3R}{5}$$

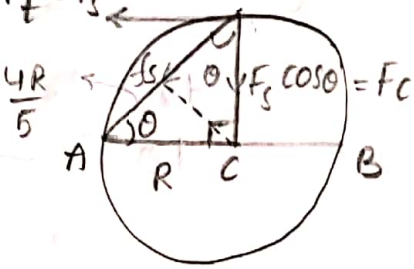
$$x = \frac{3}{5}(2) = \frac{6}{5}\ m$$

spring force,

$$F_s = Kx$$

$$= 10 \times \frac{6}{5} = 12\ N$$

$F_t = F_s \sin\theta$



Centripetal force,

$$F_c = F_s \cos\theta$$

$$\frac{mv^2}{R} = 12 \times \frac{4}{5}$$

$$\frac{800}{1000}\, v^2 = \frac{16}{5} \times 2$$

$$v^2 = 16 \times 4$$

$$\boxed{v = 8\ m/s}$$

Again,

$$F_t = F_s \sin\theta$$

$$m \cdot \frac{dv}{dt} = 12 \times \frac{3}{5}$$

$$\frac{dv}{dt} = 12 \times \frac{3}{5}$$

rate of change of speed $\leftarrow \frac{dv}{dt} = \frac{120}{5} = 24\ m/s$

Radius of Curvature

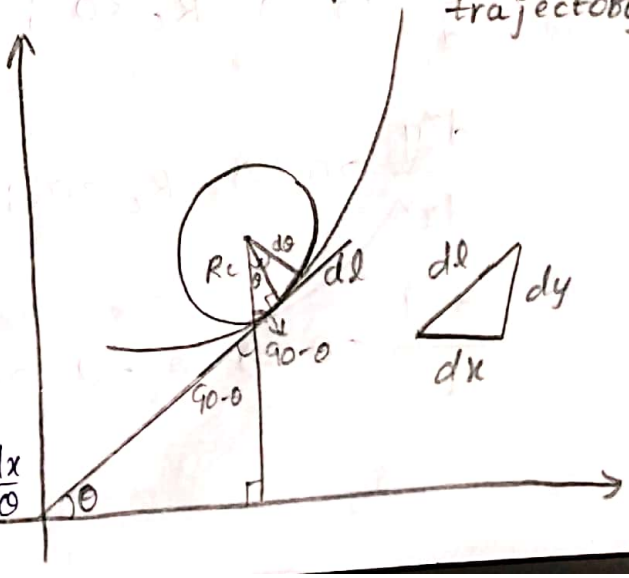
$y = f(x) \rightarrow$ eqn of trajectory.

$$dl = \sqrt{(dx)^2 + (dy)^2}$$

$$= \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$d\theta = \frac{dl}{R_c}$$

$$R_c = dl/d\theta = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, \frac{dx}{d\theta}$$



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Q slope of tangent,

$$\tan\theta = \frac{dy}{dx}$$

on differentiating,

$$\sec^2\theta \cdot \frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

$$(1+\tan^2\theta)\cdot\frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

$$\left[1+\left(\frac{dy}{dx}\right)^2\right]\cdot\frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

$$\frac{dx}{d\theta} = \frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$$

$$R_c = \frac{\sqrt{1+\left(\frac{dy}{dx}\right)^2}\left(1+\left(\frac{dy}{dx}\right)^2\right)}{\frac{d^2y}{dx^2}}$$

***

$$\boxed{R_c = \frac{\left(1+\left(\frac{dy}{dx}\right)^2\right)^{3/2}}{d^2y/dx^2}}$$

* $\frac{d^2y}{dx^2} < 0 \Rightarrow R_c < 0$; circle lies below the curve.

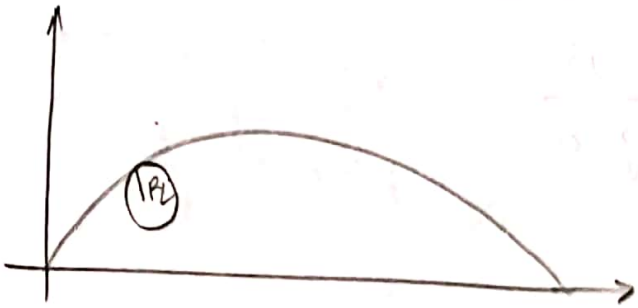
$\frac{d^2y}{dx^2} > 0 \Rightarrow R_c > 0$; circle lies above the curve.

$\frac{d^2y}{dx^2} = 0 \Rightarrow R_c \to \infty$; straight line.

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→ Equation of trajectory : $[y = 4x - x^2]$.

* find radius of ~~circular~~ curvature at (1, 3).



$y = 4x - x^2$

$\frac{dy}{dx} = 4 - 2x \Rightarrow \left.\frac{dy}{dx}\right|_{x=1} = 4 - 2 = 2$

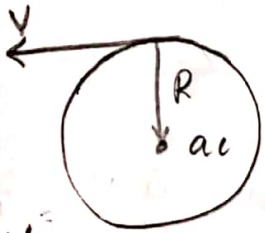
$\frac{d^2y}{dx^2} = -2$

$$R_C = \frac{(1 + (2)^2)^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

$$= \frac{5^{3/2}}{2} = \frac{5\sqrt{5}}{2}$$

II - Method :-

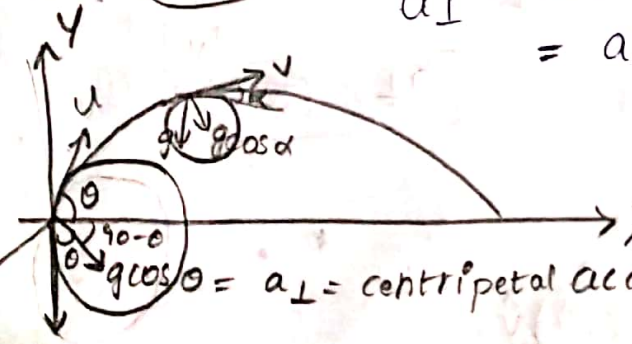
If velocity & acceleration vector are given -



$a_c = \frac{v^2}{R}$ $\quad R = \frac{v^2}{a_c} = \frac{v^2}{a_\perp}$.

$a_\perp$ ⊥ acceleration ⊥ler to velocity

$= a_t / a_r / a_N$.



$g\cos\theta = a_\perp$ = centripetal acceleration

$g\sin\theta$ ↓ tangential acceleration

$$R = \frac{u^2}{a_\perp} = \frac{u^2}{g\cos\theta}$$

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$$R_A = \frac{v^2}{g\cos\alpha}$$

$u\cos\theta = v\cos\alpha$ — (1)

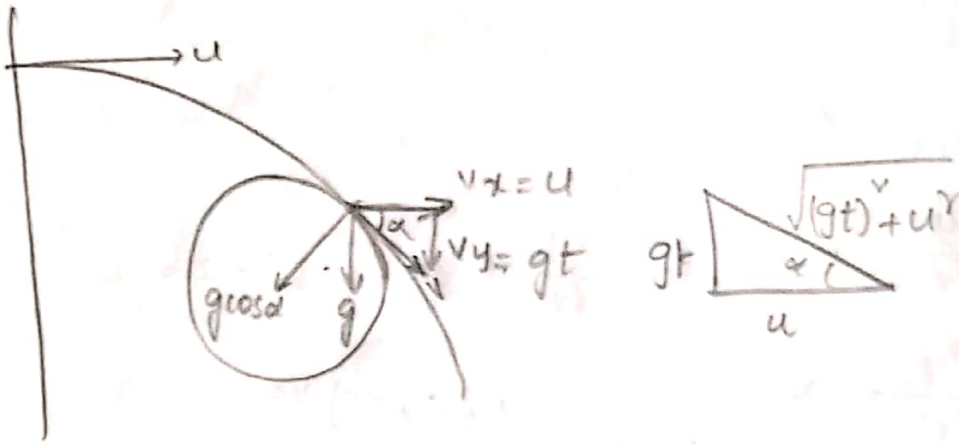
$v_x = u_x$

$v_y = u_y + a_y t$

$$\tan\alpha = \frac{v_y}{v_x}$$

$$R_A = \frac{v^2}{g\left(\frac{v_x}{v}\right)} = \frac{v^3}{g v_x} = \frac{(v_x + v_y)^{3/2}}{g\, v_x}$$

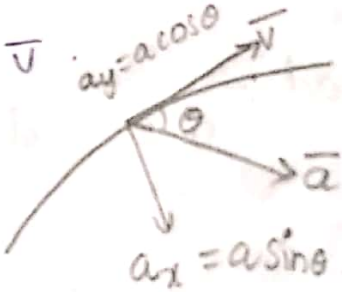
*



* let $\bar{a} = a_x\hat{\imath} + a_y\hat{\jmath}$

& $\bar{v} = v_x\hat{\imath} + v_y\hat{\jmath}$

'θ' is angle b/w $\bar{a}$ & $\bar{v}$.



we have,

$\bar{a}\cdot\bar{v} = av\cos\theta$

$$R_c = \frac{v^2}{|a\sin\theta|}$$

now,

$$a_t = a\cos\theta = \frac{\bar{a}\cdot\bar{v}}{|v|} = \bar{a}\cdot\hat{v}$$

$$\boxed{\bar{a}_t = (\bar{a}\cdot\hat{v})\hat{v}}$$

we have,

$\bar{a} = \bar{a}_t + \bar{a}_c$

$\bar{a}_c = \bar{a} - \bar{a}_t$

$\bar{a}_c = \bar{a} - (\bar{a}\cdot\hat{v})\hat{v}$.

$$\boxed{R_c = \frac{v^2}{|\bar{a} - (\bar{a}\cdot\hat{v})\hat{v}|}}$$

*

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* $\bar{a} \times \bar{v} = av\sin\theta$

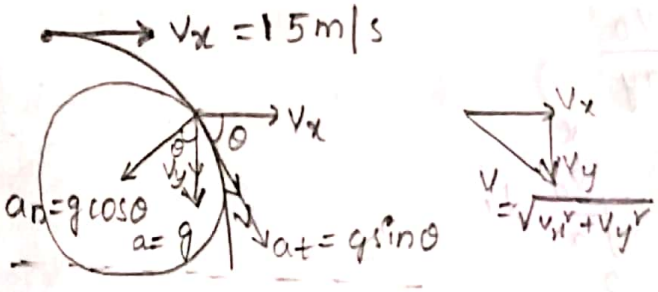
$a\sin\theta = \frac{\bar{a}\times\bar{v}}{v}$ $\quad \bar{a}\times\bar{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & 0 \\ v_x & v_y & 0 \end{vmatrix}$

$= \hat{i}(0) - \hat{j}(0) + \hat{k}(a_x v_y - v_x a_y)$

$R_c = \frac{v^2}{|a\sin\theta|} = \frac{v^3}{|\bar{a}\times\bar{v}|}$

$$\boxed{R_c = \frac{(v_x^2 + v_y^2)^{3/2}}{|a_x v_y - a_y v_x|}}$$

→ A particle is projected horizontally with velocity $v_x = 15\,m/s$. Find normal, tangential accelerations of particle at $t = 1\,sec$. take $g = 9.8\,m/s^2$



*tangential acceleration, $5.4\,m/s^2$

$$a_t = g\sin\theta = g\frac{v_y}{v} = \frac{g^2 t}{\sqrt{v_x^2 + v_y^2}} = \frac{g^2 t}{\sqrt{v_x^2 + g^2 t^2}}$$

* normal acceleration, $8.2\,m/s^2$

$$a_t = g\cos\theta = g\frac{v_x}{v} = \frac{g v_x}{\sqrt{v_x^2 + g^2 t^2}}$$

* radius of curvature,

$$R = \frac{v^2}{a_N} = \frac{v_x^2 + v_y^2}{g\cos\theta} = \frac{(v_x^2 + g^2 t^2)^{3/2}}{g v_x}$$

→ A balloon starts raising from the surface of earth with vertical component of velocity v_0. The balloon gathers a horizontal velocity $v_x =$

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ay where a is const & y is the height from the surface of the earth due to horizontal wind.

, Find eqn of trajectory of balloon.

) Find tangential, normal & total acceleration of the balloon as a function of y.

Given, $v_y = v_0$ (const)

$$\frac{dy}{dt} = v_0 \quad -①$$

& $v_x = ay$

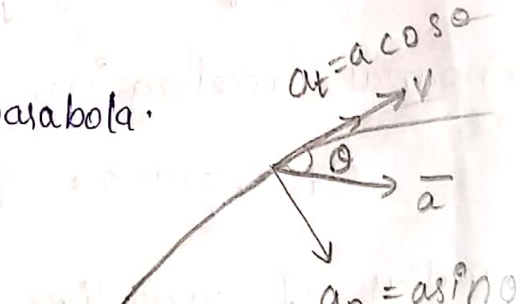
$$\frac{dx}{dt} = ay \quad -②$$

① ② $\Rightarrow$ $\frac{dy}{dx} = \frac{v_0}{ay}$

$$\int_0^y y\,dy = \int_0^x \frac{v_0}{a}\,dx$$

$$\frac{y^2}{2} = \frac{v_0 x}{a}$$

$$\boxed{y^2 = \frac{2v_0}{a}x} \rightarrow \text{parabola.}$$



let $\bar{v} = v_x\uparrow + v_y\uparrow$

$\bar{v} = ay\uparrow + v_0\uparrow \quad -③$

& $\bar{a} = a_x\uparrow + a_y\uparrow$

$$a_x = \frac{dv_x}{dt} = \frac{d(ay)}{dt} = a\frac{dy}{dt} = av_0$$

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$$a_y = \frac{dv_y}{dt} = \frac{d(v_0)}{dt} = 0$$

$$\bar{a} = a v_0 \uparrow \quad - (4)$$

now, $$a_t = a\cos\theta = \frac{\bar{a}\cdot\bar{v}}{|v|} = \frac{a^r y}{\sqrt{(ay)^r + v_0^r}}$$

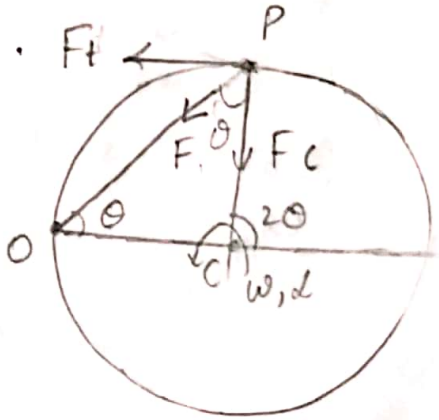
$$a_N = a\sin\theta = \frac{\bar{a}\times\bar{v}}{|v|} = \frac{a v_0^r}{\sqrt{(ay)^r + v_0^r}}$$

* Total acceleration,

$$|a| = \sqrt{a_t^r + a_N^r} = \frac{\sqrt{(a^r y)^r + (a v_0^r)^r}}{\sqrt{(ay)^r + (v_0)^r}} = \frac{a\sqrt{a^r y^r + v_0^r}}{\sqrt{(ay)^r + v_0^r}}$$

$|a| = a$

→ Particle p is moving on a circle under the action of only one force acting always towards the fixed point O on the circumference. Find the ratio of $\frac{d^r\theta}{dt^r}$ & $\left(\frac{d\theta}{dt}\right)^r$.



$$F_c = F\cos\theta = mR\omega^r$$

$$\omega^r = \frac{F\cos\theta}{mR}$$

now, $\omega = \frac{d}{dt}(2\theta)$

$= 2\frac{d\theta}{dt}$

$$2\frac{d\theta}{dt} = \left(\frac{F\cos\theta}{mR}\right)^{1/2}$$

$$\frac{d\theta}{dt} = \frac{1}{2}\left(\frac{F\cos\theta}{mR}\right)^{1/2} \quad - (1)$$

$$F_t = F\sin\theta = mr\alpha \Rightarrow \alpha = \frac{F\sin\theta}{mR}$$

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Here, $\alpha = \frac{d\omega}{dt} = \frac{d}{dt}\left(2\frac{d\theta}{dt}\right) = 2\frac{d^2\theta}{dt^2}$

$$2\frac{d^2\theta}{dt^2} = \frac{F\sin\theta}{mR}$$

$$\frac{d^2\theta}{dt^2} = \frac{F\sin\theta}{2mR} \quad -(2)$$

$$\frac{d^2\theta}{dt^2} : \left(\frac{d\theta}{dt}\right)^2 = \frac{F\sin\theta}{2mR} : \frac{1}{2}\frac{F\cos\theta}{mR} = \frac{\sin\theta}{\frac{1}{2}\cos\theta}$$

ratio = $2\tan\theta$.

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